

Linear Independence, Basis, and Dimension(线性无关, 基, 维数)

Lecture 9

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Independence, Basis, and Dimension

- 1 Independence
- 2 Basis
- 3 Dimension
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Example

Consider the following system again:

$$Ax = b \text{ is } \begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- (a) The coefficient matrix has three rows and four columns, but the third row is only a combination of the first two. After elimination it becomes a zero row. It has no effect on the homogeneous problem $Ax = 0$.
- (b) The four columns also fail to be independent, and the column space degenerates into a two dimensional plane.
- (c) By themselves, the numbers m and n of an $m \times n$ matrix give an incomplete picture of the true size of a linear system.

Rank

- The important number that is beginning to emerge (the true size) is the rank r .
- The rank was introduced as the number of pivots in the elimination process. Equivalently, the final matrix U has r nonzero rows. This definition could be given to a computer.
- But it would be wrong to leave it there because the rank has a simple and intuitive meaning:

Definition

The rank counts the number of genuinely independent rows in the matrix A .

We want definitions that are mathematical rather than computational.

Goal

The goal of this section is to explain and use four ideas:

- (a) Linear independence or dependence
- (b) Spanning a subspace
- (c) Basis for a subspace (a set of vectors)
- (d) Dimension of a subspace (a number)

Steps

- (a) The first step is to define linear independence. Given a set of vectors $v_1, \dots, v_n$, we look at their combinations $c_1 v_1 + \dots + c_n v_n$.
- (b) The trivial combination, with all weights $c_i = 0$, obviously produces the zero vector: $0v_1 + \dots + 0v_n = 0$.
- (c) The question is whether this is the only way to produce zero. If so, the vectors are independent.
- (d) If any other combination of the vectors gives zero, they are **dependent**.

Linear Independence

Definition

Suppose

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0$$

only happens when

$$c_1 = c_2 = \cdots = c_k = 0.$$

Then the vectors $v_1, v_2, \dots, v_k$ are linearly independent. If any c 's are nonzero, the v 's are linearly dependent. One vector is a combination of the others.

Remarks

- ① Linear dependence is easy to visualize in three-dimensional space, when all vectors go out from the origin.
- ② Two vectors are dependent if they lie on the same line.
- ③ Three vectors are dependent if they lie in the same plane.
- ④ A random choice of three vectors, without any special accident, should produce linear independence. Four vectors are always linearly dependent in $\mathbb{R}^3$.

Examples

- ① **Example 1** If $v_1 = \text{zero vector}$; then the set is linearly dependent. We may choose $c_1 = 3$ and all other $c_i = 0$; this is a nontrivial combination that produces zero.
- ② **Example 2** The columns of the matrix

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & 3 & 0 \end{bmatrix}$$

are linearly dependent, since the second column is three times the first. The combination of columns with weights $-3, 1, 0, 0$ gives a column of zeros. The rows are also linearly dependent; row 3 is two times row 2 minus five times row 1. (This is the same as the combination of b_1, b_2, b_3 that had to vanish on the right-hand side in order for $Ax = b$ to be consistent. Unless $b_3 - 2b_2 + 5b_1 = 0$, the third equation would not become $0 = 0$.)

Example

Example

Example 3 The columns of this triangular matrix are linearly independent:

$$\begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 2 \end{bmatrix}.$$

Look for a combination of the columns that makes zero: Solve $Ac = 0$. We have to show that c_1, c_2, c_3 are all forced to be zero. The only combination to produce the zero vector is the trivial combination. The nullspace of A contains only the zero vector $c_1 = c_2 = c_3 = 0$.

Theorem

Theorem

The columns of A are independent exactly when $N(A) = \{\text{zero vector}\}$.

A similar reasoning applies to the rows of A , which are also independent.

Independence

- The nonzero rows of any echelon matrix U must be independent.
- Furthermore, if we pick out the columns that contain the pivots.
If A has been converted to its Row Echelon Form U , then the columns with pivots of U are guaranteed to be independent.

The general rule is this:

Theorem

The r nonzero rows of an echelon matrix U and a reduced matrix R are linearly independent. So are the r columns that contain pivots.

Example 4 The columns $e_1, e_2, \dots, e_n$ of the n by n identity matrix are independent. Most sets of four vectors in $\mathbb{R}^4$ are independent. Those e 's might be the safest.

Linear Independence

- To check any set of vectors $v_1, v_2, \dots, v_n$ for independence, put them in the columns of A .
- Then solve the system $Ac = 0$; the vectors are dependent if there is a solution other than $c = 0$.
- With no free variables (rank n), there is no nullspace except $c = 0$; the vectors are independent.
- If the rank is less than n , at least one free variable can be nonzero and the columns are dependent.

One case has special importance:

Theorem

A set of n vectors in $\mathbb{R}^m$ must be linearly dependent if $n > m$.

Example 5

Example 5 These three columns in $\mathbb{R}^2$ cannot be independent.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}.$$

- To find the combination of the columns producing zero we solve $Ac = 0$:

$$A \rightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

- If we give the value 1 to the free variable c_3 , then back-substitution in $Uc = 0$ gives $c_2 = -1, c_1 = 1$.
- With these three weights, the first column minus the second plus the third equals zero: Dependence.

Spanning a subspace

Now we define what it means for a set of vectors to span a space. The column space of A is spanned by the columns. **Their combinations produce the whole space.**

Definition

If a vector space V consists of all linear combinations of $w_1, w_2, w_3, \dots, w_l$, then these vectors span the space. Every vector v in V is some combination of the w 's:

Every v comes from w 's

$$v = c_1 w_1 + \dots + c_l w_l$$

for some coefficients c_i .

Remarks

It is permitted that a different combination of w 's could give the same vector v . The c 's need not be unique, because the spanning set might be excessively large—it could include the zero vector, or even all vectors.

Examples

Example

Example 6 The vectors $w_1 = (1, 0, 0)$, $w_2 = (0, 1, 0)$, and $w_3 = (-2, 0, 0)$ span a plane (the xy plane) in $\mathbb{R}^3$. The first two also span this plane, whereas w_1 and w_3 span only a line.

Example

Example 7

- The column space of A is exactly **the space that is spanned by its columns**. The row space is spanned by the rows. The definition is made to order. Multiplying A by any x gives a combination of the columns; it is a vector Ax in the column space.
- The coordinate vectors $e_1, e_2, \dots, e_n$ coming from the identity matrix span $\mathbb{R}^n$. But the columns of other matrices also span $\mathbb{R}^n$.

Basis

To decide if b is a combination of the columns, we try to solve $Ax = b$. To decide if the columns are independent, we solve $Ax = 0$. **Spanning involves the column space, and independence involves the nullspace.** The coordinate vectors $e_1, e_2, \dots, e_n$ span $\mathbb{R}^n$ and they are linearly independent. Roughly speaking, **no vectors in the set are wasted.** This leads to the crucial idea of a basis:

Definition

A basis for V is a sequence of vectors having two properties at once:

1. The vectors are linearly independent(not too many vectors).
2. They span the space V (not too few vectors).

Remarks

Remarks:

- There is one and only one way to write v as a combination of the basis vectors. Why? Can you prove it?
- A vector space has infinitely many different bases. Which one is the best?

Figure 2.4

Example 8

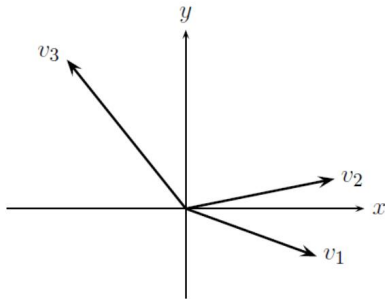


Figure 2.4: A spanning set v_1, v_2, v_3 . Bases v_1, v_2 and v_1, v_3 and v_2, v_3 .

Examples

The xy -plane in Figure 2.4 is just $\mathbb{R}^2$. The vector v_1 by itself is linearly independent, but it fails to span $\mathbb{R}^2$. The three vectors v_1, v_2, v_3 certainly span $\mathbb{R}^2$, but are not independent. Any two of these vectors, say, v_1 and v_2 , have both properties—they span, and they are independent. So they form a basis. Notice again that a vector space does not have unique basis.

Example 9

Example 9 These four columns span the column space of U , but they are not independent:

$$\text{Echelon matrix } U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are many possibilities for a basis, but we propose a specific choice: **The columns that contain pivots are a basis for the column space.**
- These columns are independent, they can span the column space.
- $C(U)$ is not the same as the column space of A , $C(A)$ before elimination—but the number of independent columns didn't change.
- To summarize: The columns of any matrix span its column space.

Dimension

A space has infinitely many different bases, but there is something common to all of these choices.

Definition

Any two bases for a vector space V contain the same number of vectors. This number, which is shared by all bases and expresses the number of “degrees of freedom” of the space, is the **dimension** of V .

Here is our first big theorem in linear algebra:

Theorem

If $v_1, v_2, \dots, v_m$ and $w_1, w_2, \dots, w_n$ are both bases for the same vector space, then $m = n$. The number of vectors is the same.

The dimension of a space is the number of vectors in every basis.

Proof.

Suppose there are more w 's than v 's ($n > m$). We will arrive at a contradiction. Since the v 's form a basis, they must span the space. Every w_j can be written as a combination of the v 's: If $w_1 = a_{11}v_1 + \cdots + a_{m1}v_m$, this is the first column of a matrix multiplication VA :

$$W = [w_1 \ w_2 \ \cdots \ w_n] = [v_1 \ v_2 \ \cdots \ v_m] \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = VA$$

we don't know each a_{ij} , but we know the shape of A (it is m by n). The second vector w_2 is also a combination of the v 's. The coefficients in that combination fill the second column of A . The key is that A has a row for every v and a column for every w . A is short, wide matrix, since $n > m$. There is a nonzero solution to $Ax = 0$. Then $VAx = 0$ which is $Wx = 0$. A combination of the w 's gives zero! The w 's could not be a basis—so we cannot have $n > m$. If $m > n$ we exchange the v 's and w 's and repeat the same steps. The only way to avoid a contradiction is to have $m = n$. □

Maximal independent set; minimal spanning set

Remark: In a subspace of dimension k , no set of more than k vectors can be independent, and no set of fewer than k vectors can span the space.

Theorem

Any linearly independent set in V can be extended to a basis, by adding more vectors if necessary. Any spanning set in V can be reduced to a basis, by discarding vectors if necessary.

Remarks:

- A basis is a maximal independent set. It cannot be made larger without losing independence.
- A basis is also a minimal spanning set. It cannot be made smaller and still span the space.

Two More Examples

Example

Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbb{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?

Example

Example

Decide whether or not the following vectors are linearly independent, by solving $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

Decide also if they span $\mathbb{R}^4$, by trying to solve

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Final Note

- 1 You must notice that the word “dimensional” is used in two different ways.
- 2 We speak about a four-dimensional vector, meaning a vector in $\mathbb{R}^4$.
- 3 Now we have defined a four-dimensional subspace; an example is the set of vectors in $\mathbb{R}^6$ whose first and last components are zero.
- 4 The members of this four-dimensional subspace are six-dimensional vectors like $(0, 5, 1, 3, 4, 0)^T$.
- 5 **NEVER** use the terms “basis of a matrix” or “rank of a space”.
- 6 The dimension of the column space is equal to the rank of the matrix.

Homework Assignment 9

2.3: 1, 2, 6, 7, 12, 13, 16, 24, 33, 37.